DEVIATHON 2.0

National Level Hackathon | GLA University, Mathura

GLA UNIVERSITY
MATHURA | GR. NOIDA
Recognised by UGC Under Section 2(f)
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TITLE PAGE

- **Problem Statement Title –** Proctoring that does not punish a shared room
- **PS Number –** *[enter PS number]*
- **Organisation –** Integraminds
- **Theme / Track –** Mission Bharat / Software
- **Team Name –** *[enter team name]* **Team ID –** *[enter team ID]*
- **Team Member Names –** *[enter names of all members]*
- **Contact –** *[team lead e-mail · phone]*



FairProctor — flag the cheating, not the room

[Team Name]

PROBLEM UNDERSTANDING

Today's proctoring flags the room — and misses the real cheating

120–390 alerts / hour
from a flag-everything proctor on our own shared-room videos — none of it cheating

0 alerts
for the most common real method: a second phone below the camera

Who pays

Candidates from small towns
rejected for their room, never told why

Reviewers
too many flags to watch — honest fail, cheaters pass

Companies and universities
results they cannot trust

Existing tools

Rule-based flags	Face / gaze AI	Live proctors
punish dim rooms, phones, 2G	blind at 2G, needs a GPU	too costly to scale

→ **Gap:** nobody separates the room from the behaviour

We must also: score on CPU · follow the DPDP Act · no face recognition · no gender / religion / caste / region inference · work on 2G phones

[Team Name]

PROPOSED SOLUTION

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DEVIATHON

FairProctor scores the behaviour, never the room

Priya

honest · 2G · dim shared room · phone held low

Old rules

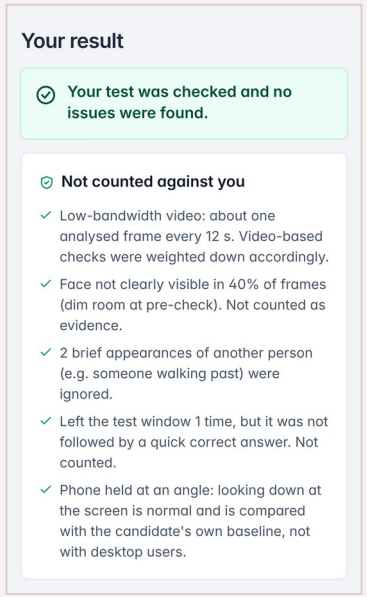
#9 of 202 — flagged

↓

FairProctor

Cleared

with 5 things it did not count



Rahul

second phone below the camera

Old rules

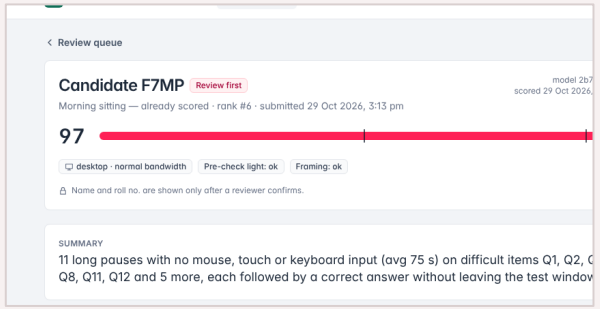
#82 of 202 — missed

↓

FairProctor

Review first · 97

11 silent pauses, each then a correct answer



- The room never adds suspicion
- Camera-free signals catch the 2nd phone
- Ranked queue, every flag with its moments
- Candidates see reasons and can appeal
- Phone pre-check fixes light first

Also: coding tests (code similarity + typing patterns) · live monitor · fairness report per sitting · English and हिन्दी

[Team Name]

TECHNICAL APPROACH

The phone sends numbers, a CPU server scores, a human decides

1 Phone

Pre-check and test. Sends timing, focus, paste length, face counts — never images

2 FastAPI + PostgreSQL

Stores events safely on 2G; JWT logins for 3 roles

3 Scoring engine (CPU)

Behaviour features + answer matching → gradient-boosted trees

4 Reviewer and candidate

Ranked queue with moments; result with reasons and appeal

Key idea: score – what the room alone would add ⇒ a dark room or a 2G line can never raise suspicion

Stack — and why

- **React + TypeScript** — phone-first, offline-safe
- **FastAPI (Python)** — same language as the ML
- **PostgreSQL** — reliable, free tier
- **scikit-learn** — CPU: 0.04 ms a session
- **On-device face detection** — no images leave the phone
- **JWT + scrypt** — secure logins, 3 roles

[Team Name]

FEASIBILITY AND VIABILITY

Already built and tested — and every risk has a plan

Working app
candidate, reviewer, admin

37 tests
pass on SQLite + PostgreSQL

6 browser suites
end-to-end

0.04 ms
to score a session on CPU

Top risks → our plan

Trained on synthetic data
→ real pilot with a sealed key; retrain on IMAP data

Honest look-alike answers
→ rare-answer weighting; a human decides; appeal

2G drops · privacy
→ offline queue; numbers not images; DPDP deletion

Plan and milestones

Done · 24–29 Oct
Scoring engine, web app, coding tests, fairness report, pilot kit

Next · before Round 2
Real pilot (10–20 people), deploy, demo video

Later
Retrain on IMAP data, per-client thresholds

Viability: CPU only — a pilot runs on free tiers; plugs into IMAP as an API or a reviewer console

[Team Name]

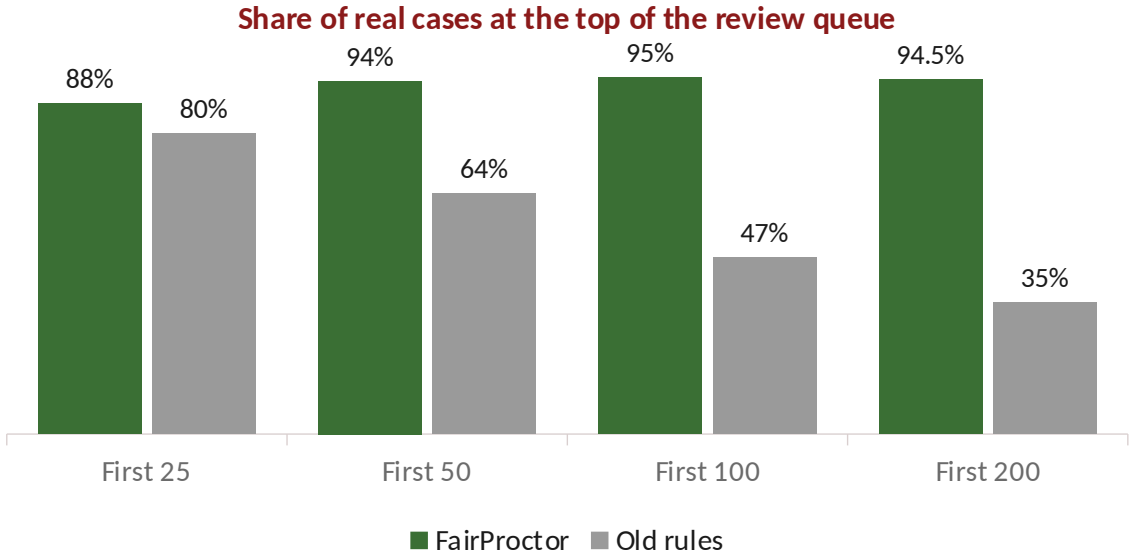
IMPACT AND BENEFITS

Real cheating rises to the top; honest 2G candidates stop paying for their room

95% vs 47%
real cases in the first 100 reviewed — ours vs old rules

55% smaller queue
and every confirmed case is still in it

+3 vs +44 pts
extra flagging of honest 2G candidates — ours vs old rules



PS "Done when" — all five met

- Queue cut by over half
- Every flag → its moments
- Works at 1 frame / 5 s
- Phone pre-check, plain words
- False flags by bandwidth

Measured out of fold on the 5,000-session labelled sample (synthetic — the IMAP data was not shared). Real-pilot numbers follow.

[Team Name]

RESEARCH AND REFERENCES

Every design choice was checked on real data

Lab-trained AI calls honest head-down “cheating” (~100%)
so we never train on lab video [2]

At 2G an eye is only ~9 pixels wide
so no gaze tracking [1]

Fast answers are normal for real learners
so timing alone never flags [3]

Walk-pasts: 120–390 alerts/hour on our videos
so brief passers-by are ignored

Audio hears speech — and cows, cookers, TV
so audio only supports [4][5]

Code and reports: *[GitHub repository link]*

References

[1] Cheating Scenario Dataset in Online Exam, Mendeley Data, doi:10.17632/mjrfmvsh7d.1 (CC BY 4.0)
[2] Students Suspicious Behaviors Detection Dataset, Mendeley Data (CC BY 4.0)
[3] Choi et al., EdNet, AIED 2020 — github.com/riiid/ednet (CC BY-NC 4.0)
[4] K. J. Piczak, ESC-50, ACM MM 2015 (CC BY-NC 3.0)
[5] Snyder, Chen, Povey, MUSAN, arXiv:1510.08484, 2015
[6] Pedregosa et al., scikit-learn, JMLR 12, 2011
[7] Schleimer, Wilkerson, Aiken, Winnowing (MOSS), SIGMOD 2003
[8] Rasch 1960; Wilson, JASA 1927; Newcombe, Stat. Med. 1998
[9] Digital Personal Data Protection Act, 2023, Govt. of India
[10] Integraminds PS, Deviathon 2.0
[11] React, TypeScript, Vite, Tailwind, FastAPI, PostgreSQL, pandas, NumPy, PyJWT, Playwright (open source)
[12] Face detection: browser FaceDetector API; MediaPipe; YuNet / OpenCV (Apache 2.0)
[13] Icons: Tabler Icons (MIT) via react-icons
[14] Screenshots: our own FairProctor prototype (demo data, people made up); chart from our results
[15] Template, Deviathon and GLA logos: organisers